

Seeing statistical regularities: Texture and pattern perception

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Abstract

The traditional view of visual texture perception emphasises its role in highlighting *discontinuities* between surfaces. Here I describe a different approach that considers how and why the visual system might capture *regularities* of texture: e.g. the “flow” of orientation structure that characterises tree bark. Such a description of an image is statistical in nature and can capture regularities that can, in turn, be used to compute properties like surface-shape. Observers are adept at estimating statistical regularities of textures, such as “mean-size” or “orientation-variance” and I review evidence bearing on which visual attributes are amenable to such statistical summary, what statistics are computable and how these computations are achieved within the visual system.

Keywords: statistics, texture, averaging, global processing, local-to-global

1. Introduction: Seeing statistics

The human visual system has evolved to effectively guide behaviour within complex natural visual environments. To achieve this goal the brain must rapidly distil a massive amount of sensory data into a compact representation that captures important image structure (Marr, 1982). Natural images are particularly rich, in part because the surfaces that populate them are often covered in markings or *texture*. This texture can be richly informative, e.g. about material composition (Kass & Witkin, 1985), but is intrinsically complex since textures are by their nature composed of a large number of individual features. One way the visual system produces a compact description of complex textures is to exploit *redundancy* (i.e. that one image-patch is not unrelated to any other patch of the same image) by characterizing attributes of the features making up the texture (such as orientation) in terms of local statistical properties (e.g. *mean-orientation*). Indeed, a useful operational definition of “visual texture” is *any image for which a statistical representation is appropriate*. To put it another way, texture is less about the image, but more about the quality of the statistic that can be computed from it (in the context of the task at-hand).

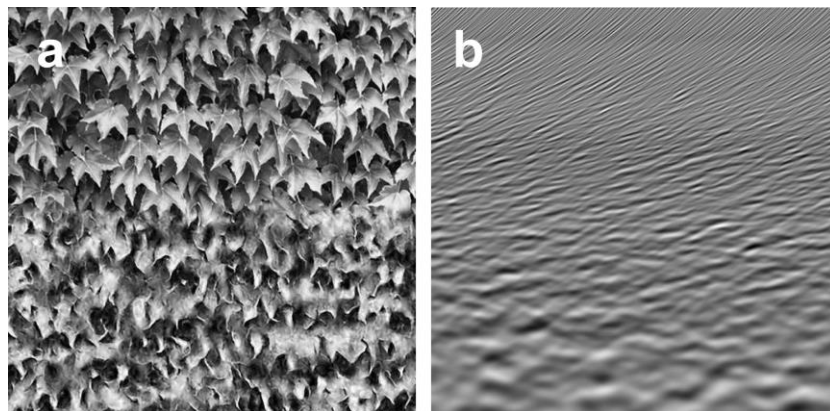


Figure 1. Statistics convey the (a) appearance and (b) shape of texture. (a) Although this image appears to be entirely natural, with scrutiny one can see that only the top half

shows real leaves. The lower half started its life as random pixel-noise that had statistical properties of the leaves imposed upon it (Portilla & Simoncelli, 1999). While statistical representations capture important properties of texture, *changes* in those statistics are also informative. For example, (b) shows a gradient defined by simultaneous changes in the mean and variance of both the size of elements and their orientation. Notice how changes in these statistics convey a vivid sense of surface-shape.

Statistics are a *sufficient* representation of natural texture in the sense that one can synthesise realistic texture based on statistical descriptions of image-features derived from histograms of e.g. grey-levels, local orientation and spatial frequency structure (Figure 1b; Portilla & Simoncelli, 1999). Since they exploit redundancy, these schemes work well on uniform regions of texture. However *changes* in statistics over space also inform our interpretation of natural scenes. Figure 1b is defined by a continuous variation in the average orientation/size and in the range of orientation/sizes present in the texture. The vivid impression of surface tilt and slant generated by this image is consistent with the visual system assuming that surface-texture is isotropic (i.e. all orientations are equally likely) so that changes in the mean and variance of orientation structure must arise from underlying changes in surface tilt and slant respectively (Malik & Rosenholtz, 1994; Witkin, 1981). Furthermore, there is evidence that these statistics drive a general and active reconstruction process that is used to resolve uncertainty about the local structure of complex scenes. Texture statistics influence the appearance of elements rendered uncertain either by visual crowding (Parkes, Lund, Angelucci, Solomon, & Morgan, 2001) or by recall within a visual memory task (Brady & Alvarez, 2011).

For the visual system to make accurate statistical descriptions it must combine information across space or time and in this chapter I focus exclusively on this *integration* process. This contrasts with the traditional view of texture perception that emphasises its role in the segmentation (Rosenholtz chapter, this volume) of the distinct surfaces that populate scenes, i.e. in the signalling of discontinuity - rather than continuity- of feature-properties across space.

Note that there is some confusion in the literature over “order” of texture-statistics. Bela Julesz proposed that humans use so-called first- and second-order statistics to capture differences in texture i.e. to achieve *texture-segmentation*. According to this terminology “first-order” refers to all grey-level (i.e. measured from single-pixels) statistics and “second-order” refers to all statistics of dipoles (pixel-pairs; Julesz, 1981; Julesz, Gilbert, Shepp, & Frisch, 1973). In this chapter I use “order” in the more conventional sense, i.e. the *order* of a histogram statistic where variance (for example) is a second-order statistic because it is computed on the square of the raw data. Thus, statistics of varying order can be computed on different image features such as “pixel-luminance” or “disc-size”, and here I will consider statistical representations on a “feature-by-feature” basis. Such an approach makes an implicit assumption that these features are appropriate “basis functions” for further visual processing (Feldman chapter on probabilistic features, this volume). For example, consider Figure 2b showing a texture composed of a ramp controlling the range of grey levels present. While this information is captured by second-order *luminance statistics* it is also captured by the first-order *contrast statistics*. Indeed this is a more meaningful characterisation of the structure in that it is contrast and not luminance that is the currency of visually-driven responses in the primate cortex. More specifically such a texture will lead to a change in the mean *response* (a first-order statistic) of a bank of Gabor filters which (like V1 neurons) are tuned for contrast and not luminance. This point is made by Kingdom, Hayes, and Field (2001) who argue that a basis set of spatial-frequency/orientation band-pass Gabor filters (Daugman, 1985) is appropriate since Gabors are not only a reasonable model of receptive field organization in V1 but can also generate an efficient/sparse code for natural image structure (Olshausen & Field, 2005). I will follow this approach and comment on the appropriateness of a basis function (size, orientation, etc) with respect to either specific neural mechanism or the model or the standard Gabor model of V1. Finally note that discrimination of the spatial structure of the pattern in Figure 2b cannot be achieved by

pooling filter-responses across the whole pattern (which e.g. could not distinguish a horizontal from a vertical gradient). Instead what is required is integration across space by mechanisms tuned to (confusingly) the “second-order” (here *contrast-defined*) spatial structure. Such mechanisms are linked to texture-segmentation and are considered in depth elsewhere (Rosenholtz chapter, this volume).

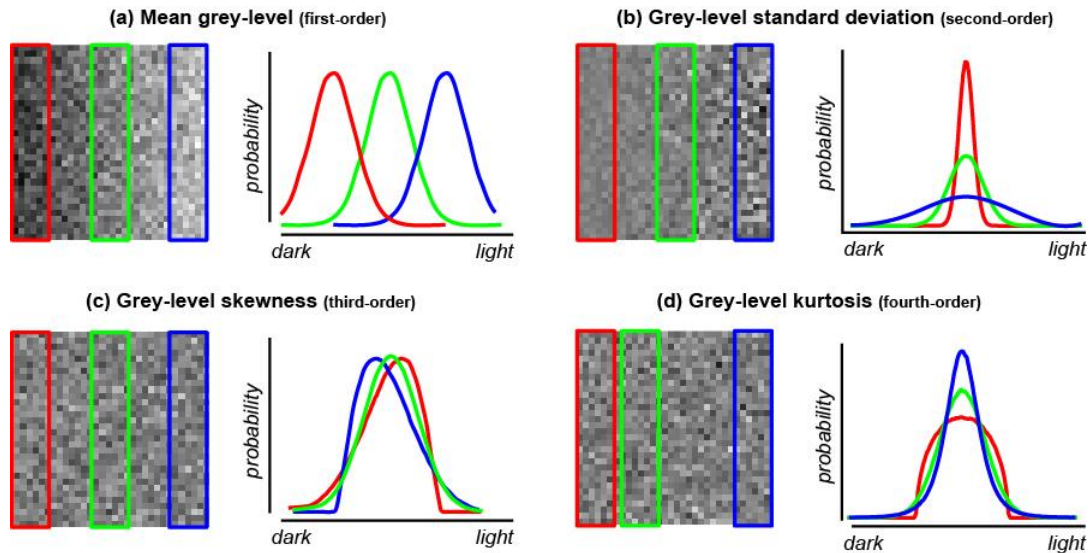


Figure 2. Noise textures made up of vertical “slices” varying in (a) first- (b) second-, (c) third- and (d) fourth-order grey-level statistics. Probability density functions for three “slices” through the image are given to the right of each texture, with curve-colour coding the slice they correspond to. Probability density functions are Pearson type VII distributions, which allow one to independently manipulate these statistical moments (http://en.wikipedia.org/wiki/Kurtosis#The_Pearson_type_VII_family). Note that the normal distribution (a,b, and green curves in c,d) is a special case of this distribution.

2. Luminance statistics

Figure 2 shows four textures containing left-to-right-variation in their (a) first- to (d) fourth-order luminance (L) statistics. Bauer (2009) reports that elements contribute to average perceived luminance (or *brightness*) in proportion to their own perceived brightness i.e. a power law $L^{0.33}$ (Stevens, 1961). However Nam and Chubb (2000) have reported that humans are near-veridical at judging the brightness of textures containing variation in luminance, with elements (broadly) contributing in proportion to their luminance. Furthermore Nam and Chubb (2000) acknowledge that while much of their data are well fit by a power function, this tends to over and underemphasise the role of the highest and lowest luminance respectively.

Different image-statistics have been proposed to capture our sensitivity to the range of luminances present (*contrast*; Figure 2b), but a good predictor of perceived contrast in complex images remains the standard deviation of grey-levels (Bex & Makous, 2002; Moulden, Kingdom, & Gatley, 1990). It should be evident from Figure 2 that the most salient changes in these noise textures are carried by the first- and second-order luminance statistics. However Chubb, Nam, Bindman, and Sperling (2007) showed that observers’ sensitivity to modulation of grey-levels is determined by “texture filters” with sensitivity to not only mean grey-level and contrast, but also to a specific type of grey-level-skewness: the presence of dark elements embedded in light backgrounds which they call “blackshot”

(Chubb, Econopouly, & Landy, 1994). Sensitivity to such skewness cannot be mediated by simple contrast-gain control¹ since the response of neurons in lateral geniculate nucleus (LGN) of cat are wholly determined by first- and second-order statistics and ignore manipulation of luminance-skew and kurtosis (Figure 2c,d; Bonin, Mante, & Carandini, 2006). Motoyoshi, Nishida, Sharan, and Adelson (2007) have suggested that grey-level skewness yields information about surface gloss with positive skew (left part of Figure 2c) being associated with darker and more glossy surfaces than skew in the opposite direction (right part of Figure 2c). However it has been argued that specular reflections (that are largely responsible for kurtosis-differences in natural scenes) have to be appropriately located with respect to underlying surface-structure in order for a percept of gloss to arise (Anderson & Kim, 2009; Kim & Anderson, 2010). This suggests that perception of material properties cannot be achieved in the absence of a structural scene-analysis. The lack of any perceptible gloss in Figure 2c is consistent with the latter view.

Kingdom et al. (2001) studied sensitivity to changes in contrast histogram statistics (variance, skew and kurtosis) by manipulating the contrast, phase and density of Gabor elements making up their textures. They report that a model observer using the distribution of wavelet/filter responses does a better job of accounting for human discrimination than raw pixel distributions.

3. Orientation statistics

In terms of spatial vision, orientation is a critical visual attribute that is made explicit at the earliest stages of representation in V1, the primary visual cortex (Hubel & Wiesel, 1962); that orientation is a property of a Gabor filter supports it being considered a reasonable basis function for studying human perception of texture statistics (Kingdom et al., 2001). Furthermore, orientation is known to be encoded in cortex using a distributed or population code, so that there are natural comparisons to be made between human coding of orientation-statistics and computational models of orientation-coding across neural populations (e.g. Deneve, Latham, & Pouget, 1999).

Miller and Sheldon (1969) used magnitude estimation to show that observers could accurately and precisely judge the average orientation of six lines spanning 20°, with each element contributing in proportion to its physical orientation. Dakin and Watt (1997) had observers classify if the mean orientation of a spatially unstructured field of elements with orientations drawn from a Gaussian distribution (e.g. Figure 3a,b) was clockwise or anti-clockwise of vertical. For elements with a standard deviation of 6° observers could judge if the mean orientation was clockwise or anti-clockwise of vertical as precisely as they could for a sine-wave grating (which contain negligible variation in orientation²). Using textures composed of two populations of elements with different means, Dakin and Watt (1997) also showed that observers rely on the mean, and not on e.g. the mode, to represent global orientation, and that observers can discern changes in the second-order statistics (orientation-variance or standard deviation - s.d.) of a texture but not in a third-order statistic (orientation-skew). Morgan, Chubb & Solomon (2008) went on to show that discrimination of changes in orientation s.d. as a function of baseline (“pedestal”) orientation s.d. follows a dipper-shaped function; i.e. best discrimination arises around a low – but demonstrably non-zero - level of orientation s.d. Such a pattern of results arises naturally from an observer basing its judgements on a second order statistic computed over orientation estimates corrupted by internal noise. However,

¹ Processes regulating neural responsivity (gain) as a function of prevailing local contrast and though to maximise information transmission in the visual pathway.

² The range of orientations present in a sine-wave grating (its orientation bandwidth) depends only on the size of the aperture the grating is presented within. In the limit, a grating of infinite size contains only one orientation. For the multi-element textures used in the averaging experiment, orientation bandwidth results from a complex interaction of element-size, element-orientation and arrangement.

Morgan *et al* found that 2/3 of their observers showed more facilitation³ than predicted by the intrinsic noise model. They speculate that this could arise from the presence of a threshold non-linear transduction of orientation variability (as it does for e.g. blur) which would serve to reduce the visibility of intrinsic noise/uncertainty and “regularise” the appearance of arrays of oriented elements.

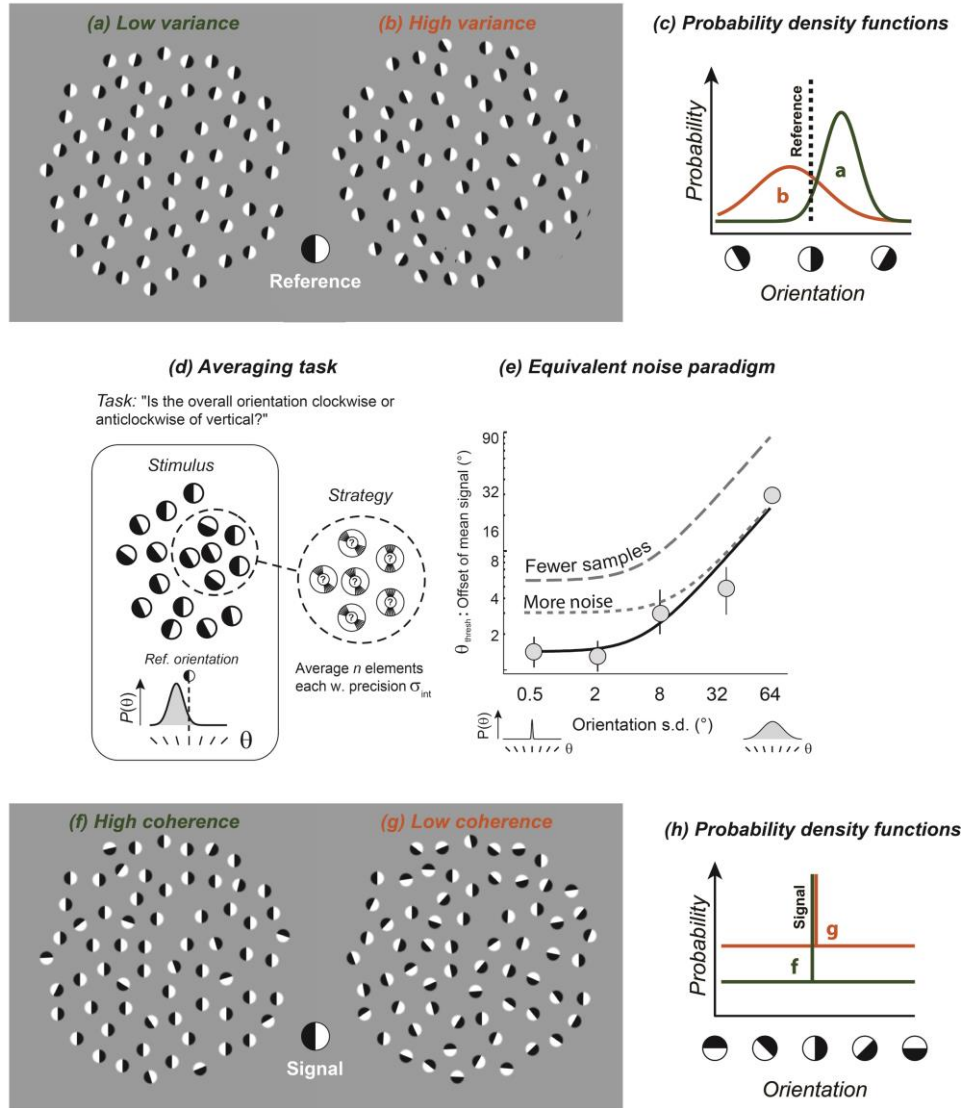


Figure 3. Probing the statistical representation of orientation. (a-b) Stimuli from a **discrimination** experiment, containing (c) differing ranges of orientation (here (a) $\sigma=6^\circ$ or (b) $\sigma=16^\circ$). (d) Observers judge if the average orientation of the elements is clockwise or anti-clockwise of a reference-orientation (here, vertical) and one experimentally determines the minimum offset of the mean (the *mean-orientation threshold*) supporting some criterion level of performance. (e) For an *equivalent noise* paradigm one measures the mean orientation thresholds with differing levels of orientation-variability and fit results with a model that yields estimates of how many samples are being averaged and how noisy each sample is. (f,g) Depicts stimuli from a **detection** experiment where observers detect the presence of a subset of elements at a single orientation (here

³ The extent to which performance improves in the presence of a low-variance pedestal.

vertical). In *coherence* paradigms one establishes the minimum proportion of elements required, here (f) 50% or (g) 12.5%, to support discrimination from randomly-oriented elements.

Such orientation statistics provide information that may support other visual tasks. Orientation variance provides an index of organization that predicts human performance on structure-versus-noise tasks (Dakin, 1999), and can be used as a criterion for selecting filter-size for texture processing (Dakin, 1997). Baldassi and Burr (2000) presented evidence that texture-orientation statistics support orientation “pop-out”. They showed that observers presented with an array of noisy oriented elements containing a single “orientation-outlier” could identify the tilt of the target element even when they couldn’t say which element was the target. Furthermore, target orientation-thresholds show a square-root dependency on the number of distracters present, suggesting that the cue used was the result of averaging target and distractor information. Observers’ ability to report the orientation of a single element presented in the periphery, and surrounded by distractors, depends on feature-spacing. When target and flanker are too closely spaced *visual crowding* arises – a phenomenon whereby observers can see that a target is present but lose detailed information about its identity (Levi, 2008). Using orientation-pop-out stimuli Parkes et al. (2001) showed that under crowded conditions observers were still able to report the average orientation (suggesting that target-information was not lost but had been combined with the flankers) and that orientation averaging does not require resolution of the individual components of the texture.

Collectively, these findings suggest that some simple global statistics computed from a pool of local orientation estimates support the detection of salient orientation structure across the visual field. But how does that process work: does pooling operate in parallel, is it spatially restricted, and is it local estimation or global-pooling that limits human performance? A qualitative comparison of orientation discrimination thresholds across conditions will not answer these questions, but rather, one needs to compare performance to an ideal observer. An equivalent noise paradigm (Figure 3a-e) involves measuring the smallest discernable change in mean orientation in the presence of difference levels of orientation variability (Figure 3a-c). Averaging performance – the threshold mean orientation offset (θ) – can then be predicted using:

$$\theta = \sqrt{\frac{\sigma_{\text{int}}^2 + \sigma_{\text{ext}}^2}{n}} \quad (1)$$

where σ_{int} is the internal noise (i.e. the observer’s effective uncertainty about the orientation of any one element), σ_{ext} the external noise (i.e. the orientation variability imposed on the stimulus), and n the effective number of samples averaged. By fitting this model to our data we can read off the global limits on performance (the effective number of samples being averaged by observers) and the local limits on performance (the precision of each estimate). This model provides an excellent account of observers’ ability to average orientation and has allowed us to show that experienced observers, confronted with N elements, judge mean orientation using a global pool of $\sim\sqrt{N}$ elements irrespective of spatial arrangement, indicating no areal limit on orientation-averaging (Dakin, 2001). Precision of local samples tends to fall as the number of elements increases at least in part due to increases in crowding (Dakin, 2001; Dakin, Bex, Cass, & Watt, 2009; Solomon, 2010), although it persists with widely spaced elements (Dakin, 2001). Solomon (2010) showed that the number of estimates pooled for orientation variance discrimination was actually higher than for mean-orientation, a finding that could perhaps arise from a strategy that weighted the contribution of elements with “outlying” orientations more heavily.

This approach assumes that observers' averaging strategy does not change with the amount of external noise added to the stimulus. Recently Allard and Cavanagh (2012) questioned this notion, reporting that the effective sample size (n) for orientation averaging changed with noise-level, which they speculate could result from a strategy-change whereby observers are less prone to pool orientations that look the same. These authors estimated sampling by taking ratios of mean-orientation discrimination thresholds collected with two different numbers of elements at the same noise level.

Combining Equation 1 with the assumption that internal noise does not change with the number of elements present, they predicted that threshold-ratios should be inversely proportional to the ratio of sampling-rates. However data from various averaging tasks (Dakin, 2001; Dakin, Mareschal, & Bex, 2005a) violate this assumption; estimates of internal/additive noise derived using Equation 1 clearly changes with the number of elements present. For this reason, estimation of sampling efficiency by computing threshold-ratios is not reasonable and Allard and Cavanagh's (2012) results are equally consistent with rises in additive noise (which Equation 1 attributes to local-orientation uncertainty) offsetting the benefits of more elements being present. What this study does do is to highlight the interesting issue of why additive noise should rise with the number of elements present on screen, especially when crowding is minimised.

Girshick, Landy, and Simoncelli (2011) examined observers' judgement of mean orientation in terms of their *precision* (i.e. threshold, variability of observers' estimate) and *accuracy* (i.e. bias, a systematic tendency to misreport the average). Observers compared the means of texture-pairs composed of orientations where either (a) both textures had high variability (a) both textures had low variability or (c) one texture had high and one low variability (this ingenious condition being designed to reveal intrinsic bias which would be matched - and so cancel - when variability levels were matched across comparisons). Authors not only measured the well-known oblique effect (lower thresholds for cardinal orientations; Appelle, 1972) in low noise conditions but also a *relative bias* effect consistent with observers generally over-reporting cardinal orientations. The idea is then that (within a Bayesian framework; Feldman chapter on Bayesian models, this volume) observers report the most likely mean orientation using not only the data to hand but also their prior experience of orientation structure (i.e. from natural scenes). Observers' performance is limited both by the noise on their read-out (the likelihood term) and their prior expectation. Using an encoder-decoder approach Girshick et al. (2011) then used variability/bias estimates to infer each observer's prior and showed that it closely matched the orientation structure of natural scenes. Consistent with this view, observers are less likely to report oblique orientations as their uncertainty rises when they become increasingly reliant on their prior expectations which are based on natural scene statistics (Tomassini, Morgan, & Solomon, 2010).

Using a coherence paradigm (Figure 1d-f; Newsome & Pare, 1988) Husk, Huang, and Hess (2012) examined orientation-processing by measuring observers' tolerance to the presence of random-oriented elements when judging overall orientation. They report that coherence thresholds were largely invariant to the contrast, spatial frequency and number of elements present (like motion coherence tasks), but that the task showed more dependency on eccentricity than motion-processing. They further showed that their data could not only reflect a "pure" integration mechanism (e.g. one computing a vector average of all signal orientation), but must also reflect the limits set by our ability to segment the signal-orientation from the noise (a process they model using overlapping spatial filters tuned to the two orientations i.e. signal-alternatives).

4. Motion statistics (direction and speed)

Reliable judgment of mean direction is possible in displays composed of elements taking random walks (with some mean direction across frames; Williams & Sekuler, 1984) or with each moving in a single directions drawn from either Gaussian or uniform random distributions (Watamaniuk, Sekuler,

& Williams, 1989). Such directional-pooling is flexible over a range of directions (Watamaniuk & Sekuler, 1992; Watamaniuk et al., 1989), operates over a large (up to 63 deg²) spatial range (consistent with large MT receptive fields) and over intervals of around 0.5s (Watamaniuk & Sekuler, 1992).

Interestingly, direction judgements are biased by the luminance content with brighter elements contributing more strongly to the perceived direction (Watamaniuk, Sekuler, & McKee, 2011). This is interesting as it suggests that the direction estimates themselves may not reflect the output of motion-tuned areas like MT which (unlike LGN or V1) exhibit little or no tuning for contrast once the stimulus is visible (Sclar, Maunsell, & Lennie, 1990). This in turn speaks to the appropriateness of element direction as a basis function for studying motion averaging. Although it is widely accepted that percept of global motion in such dot displays does reflect genuine pooling of local motion and not the operation of a motion-signalling mechanism operating at a coarse spatial scale, this is based on evidence that e.g. high-pass filtering stimuli does not reduce integration (Smith, Snowden, & Milne, 1994). A more sophisticated motion-channel that pooled coarsely across space but across a range of spatial frequencies (Bex & Dakin, 2002) might explain motion-pooling without recourse to explicit representation of individual elements. Motion coherence paradigms (analogous to Figure 3d-f) not only assume that local motion is an appropriate level of abstraction of their stimulus but that a motion coherence threshold can be meaningfully mapped onto mechanism in the absence of an ideal observer. Barlow and Tripathy's (1997) comprehensive effort to model motion coherence tasks suggests the limiting factor tends not to be a limited sampling capacity (of perfectly registered local motion) but *correspondence noise* (i.e. on registration of local motion). This is problematic for the studies that use poor performance on motion coherence tasks as an indicator of an "integration deficit" in a range of neuropsychiatric and neurodevelopmental disorders (de-Wit & Wagemans chapter, this volume).

Adapting the equivalent noise approach described for orientation we have also shown that the oblique effect for motion (poor discrimination around directions other than horizontal and vertical) is a consequence of poor processing of local motion (not reduced global pooling) and that the pattern of performance mirrors the statistical properties of motion energy in dynamic natural scenes (Dakin, Mareschal, & Bex, 2005b). Furthermore – like orientation - pooling of direction is flexible and can operate over large areas with little or no effect on the global sampling or on local uncertainty.

The standard model of motion averaging (Eqn 1) is vector summation – essentially averaging of individual (noisy) motions. However such a model fails badly on motion coherence stimuli (where it is in the observers interest to ignore a subset of "noise" directions; Dakin et al., 2005a). This flexibility – to both average over estimates and to exclude noise where appropriate - can be captured by a *maximum likelihood estimator* (MLE). In this context MLEs work by fitting a series of Gaussian templates (with profiles matched to a series of channels tuned to different directions) to simulated neural responses (subject to Poisson noise) evoked by the stimulus (Dakin et al., 2005a). The preferred direction of the best-fitting channel is the MLE direction-estimate. This model - unlike a simple vector averaging of directions - can also explain observers' ability to judge the mean direction of asymmetrical direction distributions (Webb, Ledgeway, & McGraw, 2007) better than simple vector averaging of stimulus directions. Furthermore, presence of multiplicative noise⁴ explains why sampling rate changes e.g. with the number of elements present. The MLE is a population decoder operating on combined neural responses to all of the elements present. As for any system, the more elements we add, the more information we add and so we expect the quality of our estimate of direction to improve. However, as the number of elements rise so does the overall levels of neural activity and with it the multiplicative noise. The trade-off between gains (arising from the larger

⁴ Random variability of the response of neurons in the visual pathway often rises in proportion to their mean response-level (Dean, 1981).

sample-size) and losses (because of increased noise) are captured by a power-law dependence of the effective number of elements pooled on the number of elements present (Dakin et al., 2005a). With respect to the speed of motion, observers can make an estimate of mean (rather than modal) speed over multiple elements for displays containing asymmetrical distributions of element-speed (Watamaniuk & Duchon, 1992). Speed discrimination thresholds are not greatly affected by the addition of substantial speed variation ($\mu = 7.6$, $\sigma = 1.7$ deg/sec) consistent with observers' having a high level of uncertainty about the speed of any one element of the display (Watamaniuk & Duchon, 1992). Observers can make perceptual discriminations based on the mean and variance of speed information but neither skewness nor kurtosis (Atchley & Andersen, 1995). Anecdotally, displays composed of a broad range of speeds often produce a percept not of coherent movement but of two transparent surfaces composed of either fast or slow elements. Thus performance of a mean speed task could be based on which display contains *more* fast elements. This strategy could be supported by the standard model of speed perception (where perceived speed depends on the ratio of outputs from two channels tuned to high and low temporal frequencies; e.g. Tolhurst, Sharpe, & Hart, 1973). Simple temporally-tuned channels necessarily operate on a crude spatial stimulus-representation and would predict e.g. that observers would be unable to individuate elements within moving-dot stimuli (Allik, 1992).

5. Size statistics

Looking at Figure 4 one is able to tell that the average element-size on the left and right is respectively greater or less than the size of the reference-disk in the centre. However, demonstrating that such a judgement really involves averaging has taken some time. Like orientation, early work relied on magnitude estimation to show that observers could estimate average line length (Miller & Sheldon, 1969). Ariely (2001) showed that we are better at judging the mean area of a set of disks than we are at judging the size of any member of the set. Importantly, Chong and Treisman (2003) determined what visual attribute of the disk was getting averaged by having observers adjust the size of a single disc to match the mean of two disks. They found (following Teghtsoonian, 1965) that observers pooled a size-estimate about halfway between area (A) and diameter (D): i.e. $A^{0.76}$. Chong and Treisman (2003) went on to show that observers' mean-size estimates for displays containing 12 discs were little affected by size-heterogeneity (over a ± 0.5 octave range), exposure duration, memory delays or even the shape of the probability density function for element-size. Note that when discriminating stimuli composed of disks with different mean-size there are potential confounds in terms of either overall luminance or contrast of the display (for disk- or Gabor-elements, respectively) as well as the density of element (if elements occupy the same sized regions). Chong and Treisman (2005) showed that judgements of mean element-size were unlikely to be based on such artefacts; neither mismatching density nor intermingling the two sets to be discriminated greatly impacted performance.

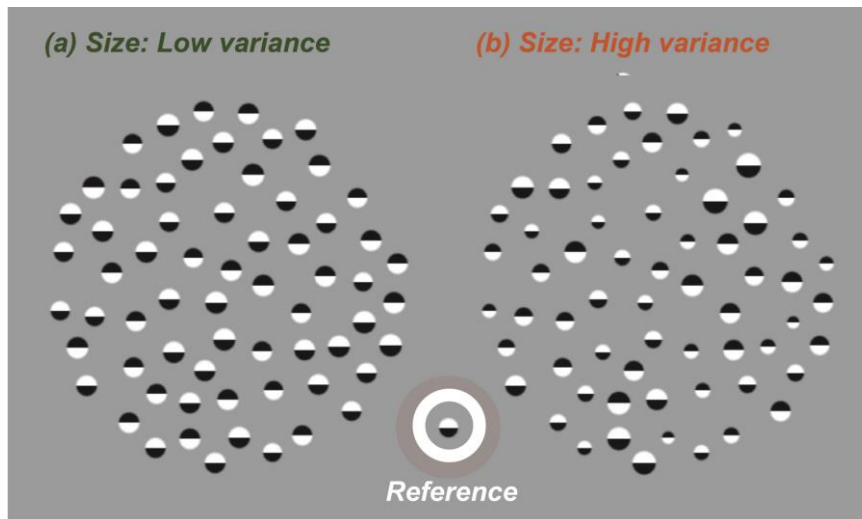


Figure 4. Even though these stimuli contain elements with either (a) low or (b) high levels of size variability, one can tell that elements are on average (a) bigger or (b) smaller than the reference.

Although carefully conducted it is difficult to draw definitive conclusions about the mechanism for size-averaging based on these early studies because of the qualitative nature of their data-analyses. Quantitative comparison of human data to the performance of an ideal observer (that averages a series of noiseless size-estimates from a subset of the elements present) led Myczek and Simons (2008) to conclude that the evidence for size-averaging was equivocal. Performance was frequently consistent with observers not averaging but rather e.g. reporting the largest element in a display. In response Chong, Joo, Emmanouil, and Treisman (2008) presented results which are intuitively difficult to reconcile with a lack of averaging (e.g. superior performance with more elements) but what hampered resolution of this debate was a failure to consistently apply a single plausible ideal observer model to a complete psychophysical data set. The ideal observer used by Myczek and Simons (2008) limited sample-size but not uncertainty about individual disk-sizes, and varied its decision rules based on the condition. To resolve this debate, Solomon, Morgan, and Chubb (2011) used an equivalent noise approach, measuring mean-size and size-variance discrimination in the presence of different levels of size-variability and modelled results using a variant on Equation 1. Their results indicate that observers can average 62-75% of elements present to judge size-variance and that (most) observers could use at least three elements when judging mean-size. Although Solomon *et al* note that performance was not substantially better than an ideal observer using the largest size present, more recent estimates of sampling for size-averaging are closer to an effective sample-size of 5 elements⁵ (Im & Halberda, 2013). This suggests that size-averaging does involve some form of pooling. Note that it is a unique benefit of equivalent noise analysis that – provided one accepts the assumptions of the ideal observer – one can remain agnostic as to the underlying psychological/neural reality of how averaging works but still definitely establish that observers perform in a manner that *effectively* involves averaging across multiple elements. Recently however, Allik, Toom, Raidvee, Averin, and Kreegipuu (2013) have presented compelling evidence that observers not only use mean-size but that this size-averaging is *compulsory* (i.e. taking place without awareness of individual sizes).

⁵ This is a corrected value based on a reported value of 7, which Allik et al (2013) point out is an over-estimate (by a factor of $\sqrt{2}$). This is because the equivalent noise model fit by Im & Halberda's (2013) does not allow for a two-interval/two-alternative forced-choice task.

There has been considerable debate in this field as to whether the number of elements present influences the observers' ability to average size. The majority of studies (Allik et al., 2013; Alvarez, 2011; Ariely, 2001; Chong & Treisman, 2005) report little gain from the addition of extra elements which has led some to conclude that this is evidence for a high-capacity parallel processor of mean size (Alvarez, 2011; Ariely, 2001). From the point of view of averaging Allik et al. (2013) points out that near-constant performance indicates a consistent drop in efficiency (i.e. sample-size divided by number of elements) and proposes a variant on the equivalent noise approach that can account for this pattern of performance.

The development of models of size-averaging that link behaviour to neural mechanisms has been limited by a general lack of knowledge about the neural code for size. As a candidate basis function for texture averaging, let us once again consider the Gabor model of V1 receptive fields. Gabor's code for spatial frequency (SF) not size. Although SF is likely a central component of the neural code for size it cannot suffice in isolation (since it confounds size with SF content). A further complication arises from the finding that the coding of size, number and density are intimately inter-connected. Randomising the size or density of elements makes it hard to judge their number and we have suggested that this is consistent with estimates of magnitude from texture (element-size, -density or -number) sharing a common mechanism possibly based on the relative response of filters tuned to different SFs (Dakin, Tibber, Greenwood, Kingdom, & Morgan, 2011). I note that such a model – like the notion that a ratio of high to low temporal-frequency tuned filters could explain speed averaging – predicts no requirement for individuation of element-sizes for successful size-averaging (Allik et al., 2013).

6. Averaging of other dimensions

Observers can discriminate differences in depth between two surfaces containing high levels of disparity noise ($\sigma=13.6$ arc min) indicating robust depth-averaging, albeit at low levels of sampling efficiency compared to other tasks (Wardle, Bex, Cass, & Alais, 2012). Like motion perception (Mareschal, Bex, & Dakin, 2008), local/internal noise limits depth averaging in the peripheral visual field (Wardle et al., 2012). de Gardelle and Summerfield (2011) looked at averaging of colour (judging “red versus blue”) and shape (“square versus circle”) as a function of the variability of the attribute and report that observers apparently assign less weight to outliers. Morgan and Glennerster (1991) showed that observers represented the location of a cloud of dots by the centroid of their individual positions with performance improving with increasing numbers of elements. Observers presented with crowded letter-like stimuli lose information in a manner consistent with features having undergone a compulsory averaging of the positions of their constituent features (Greenwood, Bex, & Dakin, 2009). As well as low-level image properties it has been shown that observers are able to make statistical summary representations of facial attributes such as emotion and gender (Haberman & Whitney, 2007) and even identity (de Fockert & Wolfenstein, 2009). Pooling of cues relating to human form even extend to pooling of biological motion (Giese chapter, this volume); observers are able to precisely judge the mean heading of crowds of point-light-walkers (Sweeny, Haroz, & Whitney, 2013).

7. Attention

Attneave (1954) argued that statistical characterisation of images could provide a compact representation of complex visual structure that can distil useful information and so reduce task demands. In this chapter I have reviewed evidence that the computation of texture statistics provide one means to achieve this goal. It has been proposed that *attention* serves essentially the same purpose, filtering relevant from irrelevant information: “... it implies withdrawal from some things in order to deal effectively with others” (James, 1890, p. 256). How then do attention and averaging

interact? Alvarez and Oliva (2009) used a change-detection task to show that simultaneous changes in local and global structure were more detectable, under conditions of high attentional load, than changes to local features alone. They argue that this is consistent with a reduction in attention to the background increasing noise in local (but less so on global) representations. However, to perform this task one had only to notice *any* change in the image, so that observers could use whatever cue reaches threshold first. Consequently, another interpretation of these findings is that global judgements are easier so that observers use them when they can. In order to determine the role of attention in averaging one must have a task where one can quantify the extent to which observers are relying on local or global information. To this end an equivalent noise paradigm (see above) has been used to assess the role of attention in averaging and, in particular, to separate its influence from that crowding (Dakin et al., 2009). Attentional load and crowding on an orientation averaging task have quite distinct effects on observers' performance. While crowding effectively made observers uncertain about the orientation of each local element, attentional restrictions limited global processing, specifically *how many elements they could effectively average*.

8. Discussion

My review suggests several commonalities between averaging of various features. Coding seems to be predominantly limited to first- and second-order statistics (sensitivity to third-order statistics in the luminance domain likely arises from the cortical basis filters being tuned for contrast, itself a second-order statistic). Computation of texture statistics generally exhibit flexibility about the spatial distribution of elements, and do not require individuation of elements. Many experimental manipulations of averaging end up influencing the *local* representation of direction and orientation (e.g. crowding, eccentricity, absolute direction/orientation) with global pooling/sampling being influenced only by attention or by the number of elements actually present. The fact that size-averaging only benefits modestly if at all from the addition of more elements is odd – and has been used to call into question whether size-averaging is possible at all. However, recent equivalent noise experiments suggest that size-averaging is possible. Further application of this technique to determine the influence of number of elements on size-averaging would allow us to determine if the lack of effect of element-number represents e.g. a trade-off between sampling-improvements and loss of local information that accompanies an increase in the number of elements.

I would sound a note of caution about the use of equivalent noise paradigms to study the human estimation of visual ensemble statistics. The two-parameter model (Equation 1) is a straightforward means of interpreting discrimination performance in terms of local/global limits on visual processing. However, this is psychophysics and the parameters such a model yields cannot guarantee that the underlying neural mechanism operates in the same manner as the ideal observer. For example, if your performance on a size-averaging task is best fit by an EN model averaging 3 elements this means you are behaving *as though* you are averaging a sample of 3 elements. In other words you could not achieve this performance using less than 3 elements. What it does *not* say is that you're necessarily averaging a series of estimates at all. As described above you could average using all the elements (corrupted by noise) or (if the sampling rate were low) just a few outlying sizes (i.e. very large or very small). Similarly, estimated internal noise - which I have termed local noise - reflect the sum of all additive noise the system is prone to. Consequently, extra noise terms can be added to the two-parameter model to capture the influence of late or decisional noise (Solomon, 2010). However, wherever noise originates, the two parameter form of this expression is still a legitimate means of estimating how much performance is being limited by an *effective precision on judgements about individual elements* and an *effective ability to pool across estimates*. I contend that this, like the psychometric function, can be treated as a compact characterisation of performance that is useful for constraining biologically plausible models of visual processing of texture statistics.

I further submit that current psychophysical data on averaging of luminance, motion, orientation, speed and perhaps size suggest a rather simple “back-pocket” model of ensemble statistical encoding. Specifically, a bank of mechanisms each pooling a set of input units (with V1-like properties) distributed over a wide range of spatial locations and spatial frequencies and with input sensitivities distributed over a Gaussian-range of the attribute of interest. Activity of each over these channels is limited by (a) effective noise on each input unit and (b) multiplicative noise on the pool and is decoded using a maximum likelihood/template-matching procedure to confer levels of resistance to uncorrelated noise (of the sort used in coherence paradigms) that a vector-averaging procedure would be unable to produce.

The cortical locus for the computation of these statistics is unknown. However it may be earlier than one might think. As well as the unexpected dependence of motion pooling on signal-luminance (indicating pooling of signals generated pre-MT) note also that while observers can average orientation signals defined by either luminance or contrast they are unable to average *across* stimulus types. This indicates that averaging happens before assignment of an abstract (i.e. cue-invariant) orientation label (Allen, Hess, Mansouri, & Dakin, 2003). As well as the issue of neural locus there are several other open questions around visual computation of summary statistics. First, what is actually getting averaged? We have seen some effort in this regard for size averaging –something between diameter and area (a “one-and-a-half-dimensional” representation?) gets averaged - but no effort has been made to separate out size from (say) spatial frequency. Building better models requires an understanding of their input. In this vein, can spatially-coarse channels of the kind described above really provide a sufficient description of images? Such a representation would predict an almost complete loss of information about individual elements under averaging. Although that does seem to happen in some circumstances the limits on the local representation have yet to be firmly established. And finally, how important are natural scenes in driving our representation of ensemble statistics other than orientation or motion?

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